What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Thalassemias

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Adapted from: Thalassemias

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Explain the pathophysiology of the mentioned anemia.
3. Interpret clinical findings to properly diagnose the mentioned anemia.
4. Outline the management of the mentioned anemia.

CASE CONNECTION

TT is a 56-year-old male seeing you for a physical. He is in excellent medical health and has no concerns. TT's physical examination is unremarkable. His blood pressure is 126/78 mm Hg. His physical form requests a complete blood count, which shows a low hemoglobin, low mean corpuscular volume, low mean cell hemoglobin concentration, and high red blood cell count.

What blood test should be ordered to confirm a diagnosis and rule out others? What management should be recommended to TT? Consider your answers as you read, and we'll revisit TT at the end of the brick.

GO TO CONCLUSION ↓

What Are Thalassemias?

Thalassemias are a group of inherited genetic blood disorders in which the body cannot make enough of the globin chains that form the backbone of the hemoglobin molecule. The two main forms of thalassemia are α-thalassemia, in which the patient cannot make enough α chains, and β-thalassemia, in which the patient cannot make enough β chains. These disorders are quantitative disorders, meaning the primary defect lies in the **amount** of globin produced; in contrast, the "structural" hemoglobinopathies (such as sickle cell anemia) are characterized by a normal amount of structurally abnormal globin chains.

Hemoglobin requires a precise balance of α and β chains. In thalassemias, the imbalance between them precipitates unbound globin chains within the red blood cell (RBC), leading to reduced production of RBC (ineffective erythropoiesis) as well as RBC lysis (hemolysis). Both lead to anemia.

Disease vs Trait

Because all the copies or alleles of globin chain genes contribute equally to the total amount of globin synthesized, the number of mutated alleles is proportional to the severity of the disease:

- Those with only one abnormal allele are usually asymptomatic.
- Those with deletion of half of the alleles have "thalassemia trait"/thalassemia minor and are usually silent carriers of the mutation. Patients may have microcytosis ± mild anemia.
- Those with more deletions have more severe anemia, thalassemia intermedia and major.

What determines the severity of thalassemia?

Epidemiology

Although less common than the other major hemoglobin disorder, sickle cell disease, thalassemias are still relatively common, seen in about 5% of people.

Thalassemias are more common in patients of Mediterranean descent and in patients from the Mediterranean, West Africa, Middle East, and Southeast Asian regions. In the United States, 85% of patients with α-thalassemia are of Asian descent, while the disorder is uncommon in Caucasians. β-Thalassemia has a high prevalence worldwide, including a 10% prevalence in malaria-endemic areas of Africa and Asia, where the trait might be protective against the malaria parasite, just as in sickle cell disease.

How Do Patients with Thalassemia Present (LO #1)?

Although the genetic defects in α- and β-thalassemia are different, the result is the same—decreased ability to make a normal quantity of hemoglobin.

Patients with thalassemia trait are asymptomatic because they generally have mild or no anemia. Patients with more severe disease describe symptoms of anemia (e.g., weakness, fatigue, pallor, dyspnea, palpitations, and chest pain). During periods of stress, such as in an infection, symptoms may worsen, and patients may note jaundice as blood cells hemolyze.

Newborns with β-thalassemia major may not develop symptoms until age 6 to 12 months. This is because fetal hemoglobin (HbF) contains α and γ globin (and does not contain β globin). HbF is still present for at least 6 months after birth and can deliver oxygen adequately. Children only become symptomatic as HbF level decreases (Figure 1).

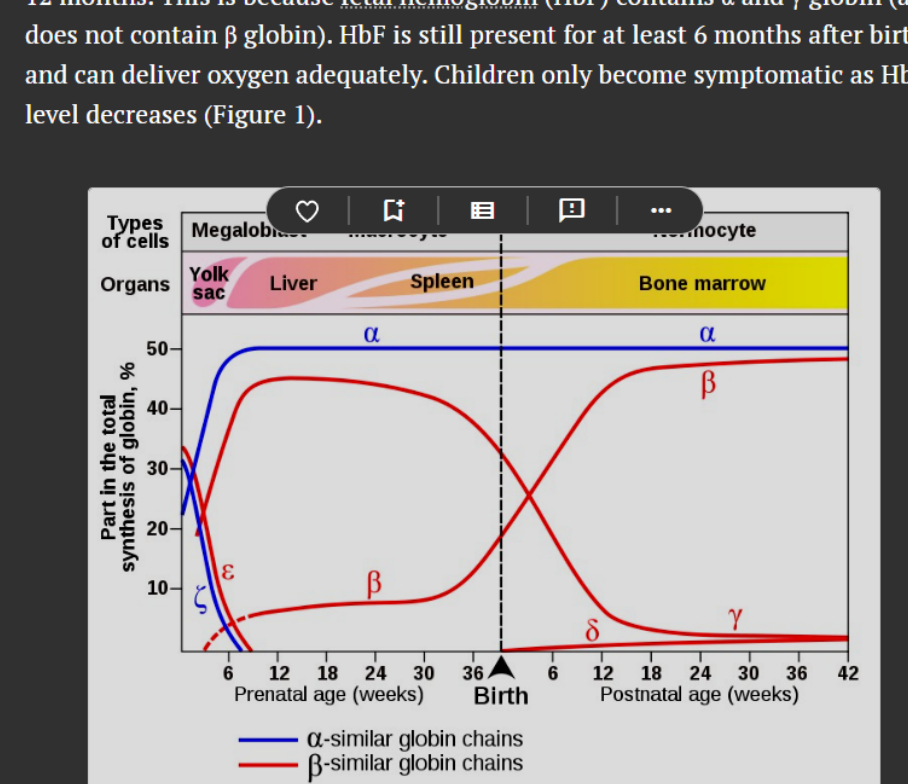

Figure 1. Globin Chain Synthesis. During fetal life, the primary hemoglobin is Hemoglobin F (α and γ globin). As the γ globin levels start to fall and β globin levels start to rise (around 6 months), β globin chain abnormalities start to display symptoms.

Many patients with thalassemia have a family history of some form of anemia. Relatives may have been falsely diagnosed with iron deficiency anemia or undergone unnecessary treatments (e.g., taking iron pills) or tests (e.g., colonoscopies looking for bleeding).

Why are neonates with thalassemia often asymptomatic at birth?

What Is the Pathophysiology of Thalassemia (LO #2)?

Thalassemia occurs because of reduced numbers of α or β globin chains because of one or more gene abnormalities or deletions.

Normal Hemoglobin

Normal adult hemoglobin, the oxygen-carrying protein in RBCs, is made of heme plus four globin chains, two α and two β (Figure 2). The ratio of α to β chains is tightly controlled at 1.0.

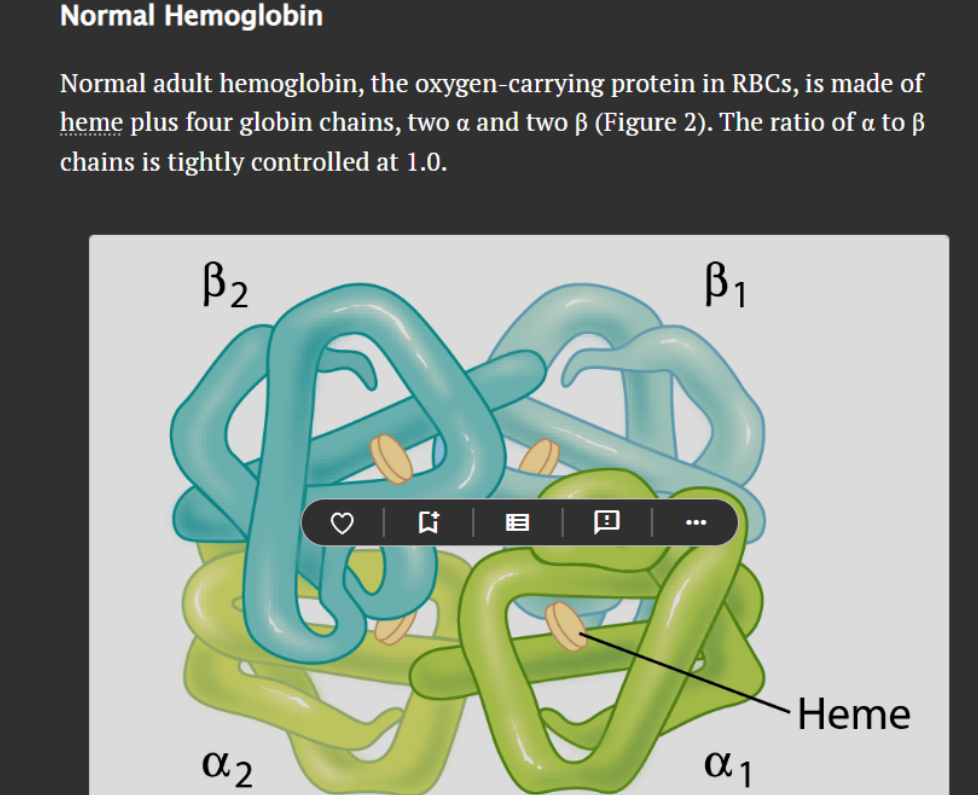

Figure 2

Other hemoglobin forms that become relevant in diagnosing thalassemias are HbA_2 with two α and two δ chains, and HbF with two α chains and two γ chains.

Anemia from Decreased α or β Chains

The anemia seen in many forms of thalassemia has several causes. In thalassemia, reduced α or β chain synthesis within the developing RBC disrupts the normal 1.0 α to β chain ratio. As a result, free α or β chains (unbound to globin) are present within the RBC cytosol. These unbound chains can precipitate within the developing RBC and damage the cell. This can lead to reduced RBC development (**ineffective erythropoiesis**) as well as early death of the RBC (increased apoptosis). Changes also occur in the RBC membrane and in intracellular water content that can make the RBC more susceptible to lysis in the circulation (**hemolysis**).

Why does decreased α or β chain production lead to anemia in thalassemia?

Genetics

Why is the synthesis of α or β chains reduced in thalassemias? This is because of a variety of **gene mutations** of the α or β chains.

α-Thalassemias are characterized by **deletions** of one or more alleles. The α globin gene has four alleles or copies, two on each chromosome 16. They are co-dominantly (equally) expressed and are needed to make the normal quantity of α chains. So, the severity of α-thalassemias depends on the number of deleted alleles.

In contrast, the problem in **β-thalassemia** is **non-deletional mutation** of one or more β chain genes, rather than gene deletions. The β chain gene has two co-dominantly expressed alleles, one on each chromosome 11. The two broad categories of mutations in β-thalassemia are those that result in no β chain production (designated β^0), and those that result in some, but not enough, β chain production (de

What is the genetic cause of α- and β-thalassemias?

α–Thalassemias

In α-thalassemia, patients have decreased α chain production. The chains that normally partner with α chains (β chains in the adult and γ chains in the fetus) end up unpaired within the RBC. As their concentration increases, these unpaired chains clump together in groups of four. In the adult, these tetramers consist of β chains called **HbH** or **H bodies**. In the fetus, they consist of γ chains called **Hb Barts**.

RBCs containing these abnormal tetramers do not develop well within the bone marrow (ineffective erythropoiesis) and are fragile and have a shorter lifespan than normal RBCs.

The severity of the patient's disease is directly related to the number of genes that have been delete

Table 1

Deleted gene	Genotype	Disease name	Description
0	αα/αα	No disease	Normal
1	αα/α-	A-Thalassemia minima (silent carrier)	No anemia, asymptomatic
2	αα/- - or α-/α-	A-Thalassemia trait	Mild anemia
3	α-/- -	HbH disease	Moderate to severe anemia
4	- -/- -	Intrauterine death (hydrops fetalis)	Extremely severe anemia, usually fatal

Deletion of One α Chain Gene. This produces the asymptomatic carrier state **α-thalassemia minima**. Because three working α chain genes are present, hemoglobin production is not significantly affected, and the patient is asymptomatic. Asymptomatic mild microcytosis without anemia or hypochromia on the peripheral blood smear can sometimes be seen, which is how it is usually diagnosed.

Deletion of Two α Chain Genes. This results in **α-thalassemia trait**. The patient has two working α chain genes, so the anemia is usually mild and occurs mostly after stressors like surgery or infection.

Deletion of Three α Chain Genes. This results in **hemoglobin H (HbH) disease.** Patients have a hard time making hemoglobin because they only have one working α chain gene. To make matters worse, HbH is unable to carry oxygen and is insolub using the RBCs to be removed from the circulation earlier than their normal lifespan.

Deletion of All Four α Chain Genes. This usually causes **intrauterine death** (**hydrops fetalis**). Fetuses with this genotype have extremely severe anemia because they cannot produce α chains. The heart attempts to compensate by beating faster and more forcefully. However, it eventually fails, and the fetus develops severe, diffuse edema, which may lead to intrauterine demise. If they survive, patients may need lifelong transfusions, gene therapy, or bone marrow transplantation.

What are H bodies?

β–Thalassemia

In β-thalassemia, patients have decreased β chain production and decreased hemoglobin with varying degrees of anemia.

Table 2 demonstrates the spectrum of disease depending on the genetics.

Table 2

Mutated gene	Genotype	Disease name	Description
0	β/β	No disease	Normal
1	β/β^0 or β/β^+	B-Thalassemia trait (minor)	Mild anemia
2	β^+/β^0 or β^+/β^+	Non–transfusion dependent thalassemia (aka thalassemia intermedia)	Moderate to severe anemia
2	β^0/β^0	Transfusion-dependent thalassemia (aka thalassemia major)	Severe, life-threatening anemia

One Mutated β Chain Gene. Patients with only one mutated β chain gene (either β^0 or β^+) have **β-thalassemia trait** (aka, β-thalassemia minor). With one normal β chain gene, β chain production is only slightly decreased, resulting in mild anemia.

Two Mutated β Chain Genes. These patients have **non–transfusion dependent thalassemia** (previously called β-thalassemia intermedia). They can make at least some β chains and have moderate anemia, but transfusions are usually not needed.

No β Chains. Patients who cannot make any β chains have **transfusion-dependent thalassemia** (previously called β-thalassemia major). This results in very severe, often l quent transfusions (hence the new name). It is usually fatal by age 5 years unless the patient can get a stem cell transplant or early intervention.

CLINICAL CORRELATION

Thalassemias and sickle cell disease can coexist. Sickle cell β-thalassemia patients have a single HbS (sickle cell) mutation in one β chain gene and a β-thalassemia mutation in the other chain. The presentations are variable but tend to feature anemia that is mild to moderate and less severe than full sickle cell anemia (HbS/HbS mutation).

Complications

Many patients with severe forms of thalassemia have increased RBC turnover and undergo frequent blood transfusions, which increase their risk of iron overload, or **secondary hemochromatosis**. This leads to iron deposition in the heart, liver, and endocrine organs and complications such as heart failure, arrhythmias, and liver disease.

How Do We Diagnose Thalassemia (LO #3)?

Although the diagnosis may be suggested by patient and family history, physical exam, and complete blood count (CBC), confirmation requires genetic testing or hemoglobin analysis.

Physical Examinati

As in any anemia, patients may have pallor (paleness) or tachycardia if the anemia is severe.

Due to ineffective erythropoiesis in the bone marrow, patients with severe thalassemia may try to make RBCs elsewhere (extramedullary hematopoiesis), especially the spleen and liver. This leads to findings of **hepatosplenomegaly**. Other sites of extramedullary erythropoiesis are the bones of the skull and face, which may result in some findings seen mostly in children with untreated disease, such as a depressed nasal bridge, maxillary hypertrophy, prominent malar eminence, and exposed upper teeth, in addition to the frontal bossing shown in Figure 3.

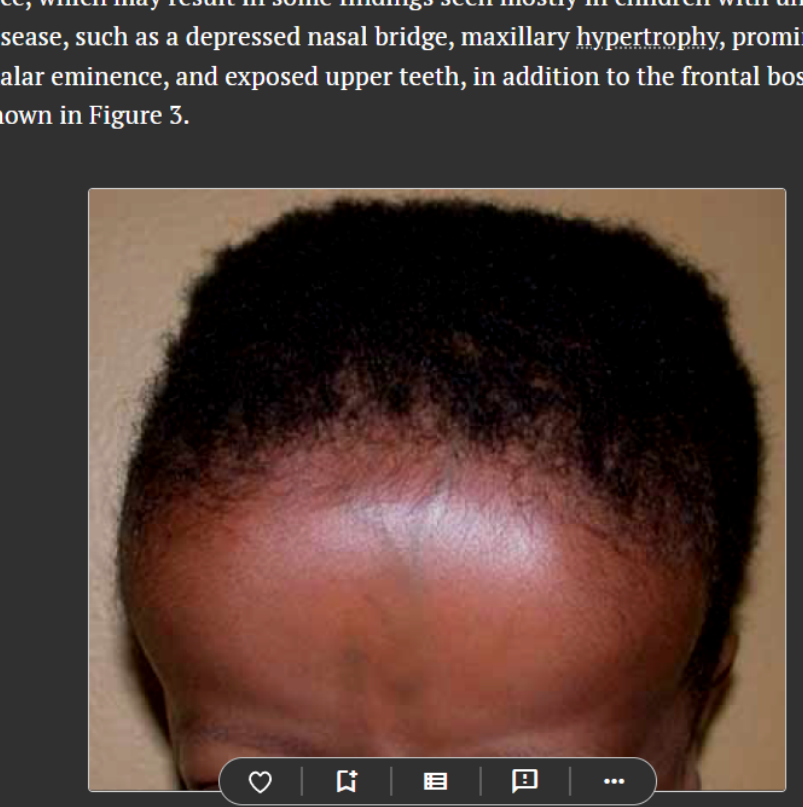

Figure 3

What is the most common physical manifestation of extramedullary hematopoiesis?

Laboratory Tests

The diagnosis of thalassemia may be suspected based on family history, the presence of anemia, and hepatosplenomegaly, but genetic testing or hemoglobin analysis is necessary to determine the precise disorder.

Complete Blood Count. The CBC in a patient with either α- or β-thalassemia typically shows low hemoglobin and hematocrit (anemia) and low mean cell volume (microcytosis), similar to the more common iron-deficiency anemia. The degree of anemia and microcytosis varies according to the severity of the genetic defect.

Interestingly, although the hemoglobin is low in mild thalassemia, the **RBC count is usually normal or increased,** unlike in iron deficiency anemia in which the RBC count is low—this is an important distinguishing factor to remember. This phenomenon can be a clue in the CBC, the so-called **"thalassemic indices."** The RBC count drops in severe thalassemias, however.

Table 3: Clues from CBC- Iron Deficiency vs Thalassemia

Iron Deficiency	Thalassemia Minor
RBC count decreased	RBC count normal or increased
Hemoglobin proportionally more decreased than MCV	MCV proportionally more decreased than hemoglobin
RDW increased	RDW normal

Peripheral Blood Smear. Key findings on blood smear can include hypochromasia, anisocytosis (wide range of RBC size), microcytosis, and target cells, all shown in Figure 4.

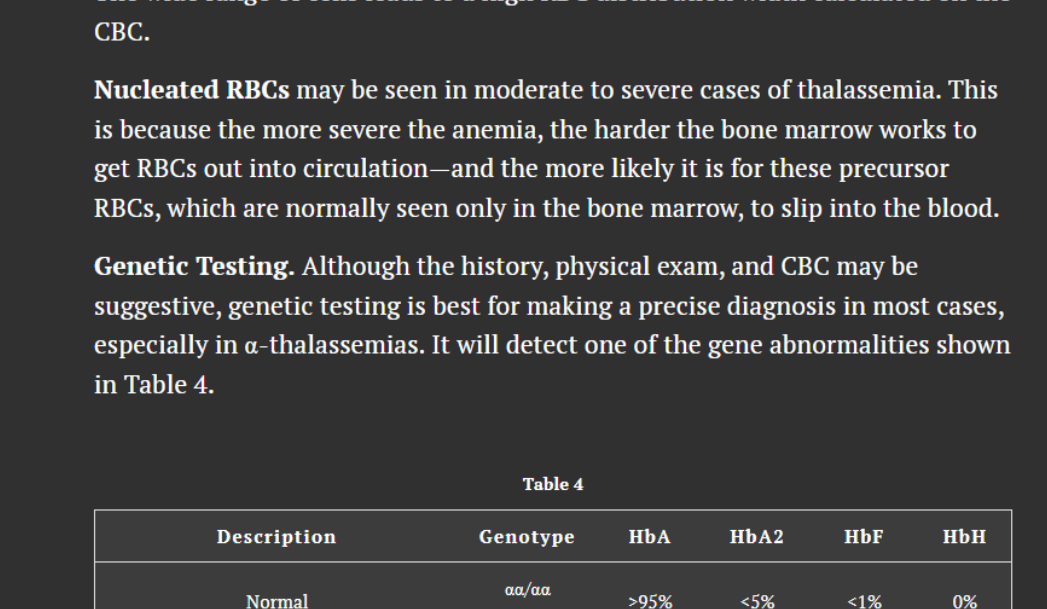

Figure 4

The wide range of cells leads to a high RBC distribution width calculated on the CBC.

Nucleated RBCs may be seen in moderate to severe cases of thalassemia. This is because the more severe the anemia, the harder the bone marrow works to get RBCs out into circulation—and the more likely it is for these precursor RBCs, which are normally seen only in the bone marrow, to slip into the blood.

Genetic Testing. Although the history, physical exam, and CBC may be suggestive, genetic testing is best for making a precise diagnosis in most cases, especially in α-thalassemias. It will detect one of the gene abnormalities shown in Table 4.

Table 4

Description	Genotype	HbA	HbA2	HbF	HbH
Normal	αα/αα β/β	>95%	<5%	<1%	0%
Silent carrier	αα/α-	Normal	Normal	Normal	0%
α-Thalassemia trait	αα/-- or α-/α-	Normal	Normal	Normal	0%
HbH disease	α-/--	Decreased	Normal	Normal	0%-30%
β-Thalassemia tr	[illegible]	[illegible]	[illegible]	Up to 5%	0%
Non-transfusion dependent thalassemia	β^+/β^0 or β^+/β^+	Decreased	Increased	Up to 50%	0%
Transfusion-dependent thalassemia	β^0/β^0	0%	Increased	Up to 95%	0%

Hemoglobin Electrophoresis. If gene analysis is not available, hemoglobin electrophoresis is often adequate to diagnose β-thalassemias (see Table 4). This test looks at the percentage of the different types of hemoglobin molecules present in the blood.

As you can see in the table, if HbA is low due to low α or β globin levels, another form of hemoglobin (eg, HbA_2 or HbF) will rise to compensate. For example, as genetic abnormalities of β-thalassemia worsen, the HbF percentage, usually only seen in fetuses and infants, increases from less than 1% to more than 95% in adults when HbA is not available.

Radiology

Hyperplasia of the bone marrow may lead to modeling deformities of the bone, such as metaphyseal expansion ("Erlenmeyer flask deformities" of long bones), as seen in Figure 5.

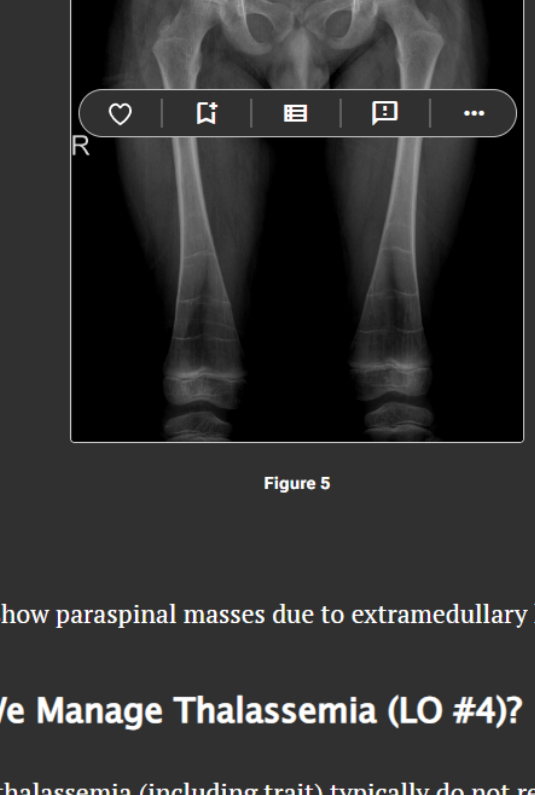

Figure 5

CT scan may show paraspinal masses due to extramedullary hematopoiesis.

How Do We Manage Thalassemia (LO #4)?

Mild cases of thalassemia (including trait) typically do not require treatment because anemia and symptoms are absent or mild.

The goals of treating anemia in individuals with thalassemia include reducing symptoms and morbidities associated with anemia (such as impaired growth and development in childhood), reducing or preventing extramedullary hematopoiesis, bone expansion, and hypersplenism, and reducing excess iron stores associated with increased intestinal iron absorption and/or transfusion.

More severe cases of thalassemia may require **chronic transfusions** to keep the patient's hemoglobin stable within an appropriate range. Patients receiving chronic transfusions must receive iron-chelating agents like deferoxamine to prevent iron overloa ometimes be reduced using luspatercept, which increases bone marrow RBC production by binding transforming growth factor-β ligands on erythroid precursor cells in the bone marrow. This leads to enhanced erythroid maturation.

Splenectomy may be appropriate for individuals with thalassemia (typically beta thalassemia) who have one or more of the following findings: severe anemia, a dramatic increase in transfusion requirements, growth retardation, hypersplenism leading to other cytopenias, symptomatic splenomegaly, and splenic infarction or splenic vein thrombosis.

CLINICAL CORRELATION

Very severe cases of thalassemia may be treated definitively with **stem cell transplant**. This treatment is reserved for the most severe cases because its benefit in these patients outweighs the risk of transplant-related complications such as immunosuppression and graft vs host disease. **Gene therapy** is increasingly being used instead in these patients and is likely to become the best therapy option.

What is the main risk of chronic transfusion therapy, and how is it

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TT, what blood test should be ordered to confirm this diagnosis and rule out others? What management is recommended?

TT has microcytic hypochromic anemia with a high red blood cell (RBC) count, making thalassemia the most likely diagnosis. The asymptomatic findings and mild anemia make α-thalassemia minor or β-thalassemia trait most likely. A check of his serum ferritin is needed to rule out an underlying iron deficiency, and it is normal. A hemoglobin electrophoresis shows β/β^0 and confirms β-thalassemia trait. You explain this to TT and its implication for his children. Because he is asymptomatic and his hemoglobin level is not extreme, no further treatment is necessary.

Summary

- Thalassemia is a group of diseases characterized by anemia due to decreased globin chain production.
- The two main forms are α-thalassemia, in which the patient cannot make enough α chains, and β-thalassemia, in which the patient cannot make enough β chains.
- Symptoms of thalassemia depend on the severity of anemia, which in turn depends on the genotype.
- Symptoms range from asymptomatic to severe, life-threatening, transfusion-dependent anemia.
- α-Thalassemia is caused by deletion of one or more of the four α chain genes.
- β-Thalassemia is caused by mutation of one or both of the β chain genes.
- Patients with thalassemia are anemic because of diminished production of α or β chains, which precipitate as tetramers within the developing red blood cell (RBC), leading to fragility and poor RBC development.
- The complete blood cell count shows microcytic, hypochromic anemia.
- RBC count is normal or increased in mild thalassemia.
- The blood smear may show anisocytosis, target cells, and nucleated RBCs in more severe cases.
- Genetic analysis and hemoglobin electrophoresis are used to make a firm diagnosis.
- Treatment of thalassemia depends on the severity of the anemia and can range from no treatment to blood transfusions to stem cell transplant and gene therapy.
- Iron chelators such as deferoxamine are used to prevent and treat iron overload from multiple transfusions.

Review Questions

1. Which of the following genetic defects is present in hemoglobin H disease?

A. ○ Deletion o
B. ● Deletion of three α chain genes
C. ○ Mutation of all four α chain genes
D. ○ Mutation of both β chain genes
E. ○ Mutation of one α chain gene

Hide Explanation

The correct answer is deletion of three α chain genes (B). HbH disease is a severe form of α-thalassemia in which three of the four α chain genes are deleted, and the excess β chains form a HbH tetramer. Deletion of one β chain gene (A) is incorrect, as β chain deletions are not present in α- or β-thalassemia (β-thalassemia is characterized by mutations, not deletions, in β chain genes). α-Thalassemias are characterized by deletions of α chain gene, not non-deletional mutations, which makes both mutation of all four α chain genes (C) and mutation of one α chain gene (E) incorrect. Mutation of both β chain genes (D) is present in β-thalassemia intermedia and β-thalassemia major.

2. Which of the following red blood cell indices is most elevated in severe thalassemia?

A. ○ Hemoglobin
B. ○ Mean cell hemoglobin concentration
C. ○ Mean corpuscular volume
D. ○ RBC count
E. ● RBC distribution width

Hide Explanation

The correct answer is RBC distribution width (E). The red blood cell distribution width (RDW) is a measure of anisocytosis (variation in red cell size); the higher the RDW, the greater the range of red cell sizes present. In severe thalassemia, the RDW is elevated because although all the red cells are small, their sizes vary. In mild thalassemia the RDW is low to normal. Hemoglobin (A) is decreased in thalassemia, as are the mean cell hemoglobin concentration (B) and mean corpuscular volume (C). The RBC count (D) is increased in milder thalassemias but is decreased in severe thalassemia.

3. Which of the following is the most common complication associated with chronic blood transfusions used to treat severe forms of thalassemia?

A. ○ Graft vs host disease
B. ○ Hepatitis C infection
C. ○ HIV infection
D. ○ Increase
E. ● Iron overload

Hide Explanation

The correct answer is iron overload (E). Chronic transfusions for any type of anemia (including thalassemia) carry a risk of iron overload unless iron chelators are administered. Graft vs host disease (A) is a potential complication of stem cell transplant but is not seen in normal blood transfusion therapy. Hepatitis C (B) and HIV (C) infections can both be caused by transfusion of infected blood products; however, the risk of contracting hepatitis C or HIV from a transfusion is rare in developed countries because of adequate blood screening. Increased hemolysis (D) is not a risk in patients receiving chronic transfusions, as the risk of being transfused the wrong blood type is very rare.

4. A 25-year-old male presents with easy fatigability and skin discoloration. He is chronically jaundiced and anemic since childhood, requiring transfusions for multiple episodes of severe anemia. Physical exam shows splenomegaly, frontal bossing, prominent malar eminence and upper teeth, and icterus. The complete blood count (CBC) shows a microcytic, hypochromic anemia. Hemoglobin electrophoresis shows elevated HbA_2 and F with undetectable HbA levels. What is the most likely diagnosis in this patient?

A. ○ α-Thalassemia trait
B. ○ β-Thalass
C. ○ Hemoglobin Bart disease
D. ○ Hemoglobin H disease
E. ● Transfusion-dependent thalassemia

Hide Explanation

The correct answer is transfusion-dependent thalassemia (E). This is suggested by the transfusion requirements, anemia, splenomegaly, and bossing (suggesting extramedullary hematopoiesis) with a hemoglobin electrophoresis showing no HbA. α -Thalassemia trait (A) is usually asymptomatic and would show a normal hemoglobin electrophoresis. β-Thalassemia trait (B) is also asymptomatic, and hemoglobin electrophoresis will show low levels of HbA and elevated HbA_2 and F. Hemoglobin Bart disease (C) is incompatible with life and causes intrauterine fetal demise. Hemoglobin H disease (D) causes severe symptomatic anemia, however, the levels of HbA, A_2, and F will be low while HbH levels will be elevated on electrophoresis.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of the mentioned anemia.~~
02 ☑ ~~Explain the pathophysiology of the mentioned anemia.~~
03 ☑ ~~Interpret clinical findings to properly diagnose the mentioned anemia.~~
04 ☑ ~~Outline the management of the mentioned anemia.~~

Objective Confidence: 4/4